

Firewall Configuration Report

Elevate Labs — Cybersecurity Internship | Task 4

Target System	localhost (naveen-Aspire-AS15-42)	Date	June 4, 2026
OS	Ubuntu 24.04	Assessor	Naveen V
Tool	UFW (Uncomplicated Firewall)	Status	Completed

Objective

Configure and test basic firewall rules on Ubuntu Linux using UFW to allow or block network traffic on specific ports. Document all commands used and verify rules are working correctly through connection testing.

What is UFW?

UFW (Uncomplicated Firewall) is Ubuntu's built-in firewall tool. It acts as a security barrier between your machine and the network — checking every connection against rules, blocking unauthorized access while allowing legitimate traffic.

Feature	Details
Full name	Uncomplicated Firewall
Built into	Ubuntu Linux (pre-installed)
Rule types	ALLOW, DENY, REJECT, LIMIT
IPv6 support	Yes — automatically applies rules to both IPv4 and IPv6

Steps Performed

Step 1 — Check Firewall Status

Listed existing firewall rules:

```
sudo ufw status
```

Result: Firewall active. Port 631 (CUPS) already blocked from previous task.

Step 2 — Block Port 23 (Telnet)

Added rule to block inbound Telnet traffic:

```
sudo ufw deny 23
```

Result: Rule added — IPv4 and IPv6 both added automatically.

Step 3 — Test the Rule

Tested if port 23 is actually blocked using Netcat:

```
nc -zv 10.14.218.12 23
```

Result: Connection refused — rule working correctly.

Test	Command	Result	Status
Port 23 blocked?	nc -zv 10.14.218.12 23	Connection refused	PASS
Port 631 blocked?	nc -zv 10.14.218.12 631	Connection refused	PASS

Step 4 — Allow Port 22 (SSH)

Added rule to allow SSH access:

```
sudo ufw allow 22
```

Result: Rule added — IPv4 and IPv6 both added automatically.

Step 5 — Remove Test Rules

Deleted test rules to restore original state:

```
sudo ufw delete allow 22 sudo ufw delete deny 23
```

Result: Both rules deleted. Final state — only port 631 blocked.

Final Firewall State

Port	Action	Protocol
631	DENY	IPv4 + IPv6

All Commands Used

Command	Result
sudo ufw status	Active — showed existing rules

<code>sudo ufw deny 23</code>	Rule added (IPv4 + IPv6)
<code>nc -zv 10.14.218.12 23</code>	Connection refused — PASS
<code>sudo ufw allow 22</code>	Rule added (IPv4 + IPv6)
<code>sudo ufw status</code>	Showed all 3 active rules
<code>sudo ufw delete allow 22</code>	Rule deleted
<code>sudo ufw delete deny 23</code>	Rule deleted
<code>sudo ufw status</code>	Only port 631 remains

How Firewall Filters Traffic

UFW checks every incoming connection against the rules list top to bottom. First matching rule wins — DENY blocks it, ALLOW lets it through. If no rule matches, default policy applies — all incoming blocked.

Step	What happens
1. Packet arrives	Connection attempts to reach your machine on a port
2. UFW checks rules	Looks through rules list top to bottom
3. DENY matched	Connection blocked — Connection refused
4. ALLOW matched	Connection passes through to the service
5. No match	Default policy — incoming blocked

Key Learnings

#	Learning
1	UFW automatically applies rules to both IPv4 and IPv6 with a single command
2	Deleting a rule removes it completely — not just disabled, fully gone
3	UFW does not block loopback (127.0.0.1) — only external network traffic
4	Port 23 (Telnet) must always be blocked — plain text, no encryption
5	Port 22 (SSH) allowed only when SSH server is actually installed
6	Testing with nc (Netcat) confirms rules are enforced, not just listed